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CHANGE DETECTION USING SYNTHETIC APERTURE SONAR IMAGERY WITH VARIABLE TIME INTERVALS

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Abstract: *This paper addresses automatic change detection in synthetic aperture sonar (SAS) imagery with time intervals ranging from two days up to several years. This is a vital parameter in change detection, as temporal decorrelation of the seafloor may exclude coherent processing of high frequency sonar data for survey intervals exceeding a few days. Due to the high costs of large area underwater surveys, however, relevant time intervals are significantly longer than this for many change detection applications, such as route surveys for mine counter measures (MCM), pipeline inspection and environmental monitoring.*

An incoherent processing chain for SAS change detection is described. The method can be applied for both short and long intervals. After image preprocessing, the SURF algorithm is used to extract and match feature points in the reference and repeat pass SAS images for coregistration. The reference image is then spatially transformed onto the pixels of the repeat pass image, and a difference image is produced by pixel-wise subtraction. Mutual sonar responses from e.g. rocks are suppressed in the difference image, thus enhancing the temporal changes. The difference image can be thresholded to detect changes or further processed to enhance specific response change patterns. The method is tested on data recorded in Norwegian waters with HISAS 1030 sonar mounted on HUGIN autonomous underwater vehicles (AUV). The results show that change detection significantly increases object detection performance, as permanent clutter objects are effectively filtered out, even for long survey intervals. As expected, the temporal variations in the seafloor sonar response increase with interval length, though.

Keywords: *Incoherent Change Detection, Synthetic Aperture Sonar, Repeat Pass Surveys*

1. INTRODUCTION

Change detection (CD) is the process of identifying objects or other phenomena of interest as temporal differences by observing a scene at different times. It has received widespread interest over the last decades due to a large number of applications within e.g. remote sensing, manufacturing quality control, video surveillance and medical diagnosis. As manual analysis of sequential images is tedious and prone to errors, significant efforts have been put into the development of automated systems that can compare sets of imagery and report only the changes relevant for the given application [1-3].

CD approaches can be categorized based on the data level where the temporal matching takes place: decision or image. Decision level methods match the output labels from a classifier operating independently on the new and old image data. The classifier must be generated based on explicit models or training data for the case. Changes are detected as class transitions between the two data sets. The CD accuracy is decreased by errors in the individual classifications and data association, but is otherwise unaffected by varying data acquisition conditions and background changes. On the other hand, image level methods match new and old image data, either for regions or individual pixels. Regions (image windows) can be compared by correlation or statistical measures calculated from the pixels' values. Many image level methods produce a difference image which needs interpretation to detect changes, typically by applying some statistical threshold. Their CD accuracies are limited by data coregistration errors, differences in imaging conditions (e.g. sensing geometry, noise and signal propagation) and irrelevant scene changes. Within the satellite remote sensing literature, decision and image level approaches are often referred to as supervised and unsupervised, respectively.

Mine hunting is an important underwater application of CD. Reference sonar imagery of strategic ports, inlets or sea lines of communications is recorded during route surveys, when the seafloor is assumed to be free of mines. After a new survey, mines are recognized as objects that are only present in the current imagery. CD can be the only feasible option when the seafloor clutter density is high, or when the mine's physical characteristics are unknown, as for improvised explosive devices (IEDs). Other promising CD applications include Intelligence, Surveillance and Reconnaissance (ISR), seafloor pipeline inspection and environmental monitoring of e.g. deep water coral reefs and offshore installation surroundings. Due to the high costs of large area underwater surveys, relevant time intervals may be in the order of one year or more.

Automated CD in sonar imagery is a relatively young research area, and has been focused on object detection using decision [4-8] level methods. This is due to several challenges imposed by the underwater environment regarding geographical coregistration of the data (no readily available GPS position updates), sensor trajectory control and complex signal propagation conditions. Additionally, traditional side-scan sonar (SSS) has severe limitations caused by poor and varying along track resolution. The introduction of synthetic aperture sonar (SAS) mounted on autonomous underwater vehicles (AUV) with aided inertial navigation systems has partly remedied these problems and triggered a growing research activity on image level methods [6,9-14]. The survey intervals in these studies have typically been a few days or less. The technical contribution of this paper is to investigate the feasibility of image level SAS CD for operationally realistic intervals.

2. SAS CHANGE DETECTION PROCESSING

Both detection sensitivities and operational requirements vary significantly between the CD approaches for SAS imagery.

Decision level methods detect objects or other features independently in the data from both surveys and correlate their positions to identify unmatched detections. This imposes no restrictions on the similarity of sensor trajectories or seafloor backgrounds, as long as all mutual targets are detected for both passes. Construction of the detector, however, requires that the relevant types of changes are defined in advance. Coregistration only needs to facilitate correct association of detections, so the vehicle navigation solution may suffice in areas with little clutter. Detection sensitivity is however bounded from above by the single pass detection performance, which can be poor for seafloors with heavy clutter and roughness. The method is thus best suited for benign seafloors and the detection of specific features.

Image level methods for coherent sensors like SAS can compare data either coherently (using both pixel magnitude and phase values) or incoherently (magnitude only). The former performs complex correlation of small, coregistered windows in the new and old images, and detects new objects and other changes as locations with reduced coherence (data similarity). It is the preferred method for synthetic aperture radar CD [3], and its detection sensitivity is excellent. Because the method utilizes both signal amplitude and phase, it can detect even small, subtle changes that are invisible in the magnitude SAS image. However, the operational demands are stringent [9,10,14]. The data must be coregistered to within one-tenth of the pixel size and the sensor trajectories offsets must be small enough to avoid degraded coherence due to different imaging geometry (baseline decorrelation). Further, high repeat pass coherence can only be achieved when the signal-to-noise ratio (SNR) is high in both single pass images. The method is thus vulnerable to e.g. surface multi-path signal pollution and inevitably marks sonar shadows of mutual objects as low coherence areas. Still, the main challenge for the operational use of coherent SAS CD is arguably temporal decorrelation. As the method is sensitive to the distribution of individual sonar scatterers within a resolution cell, only minute changes of the seafloor caused by waves, currents or biological activity is sufficient to reduce the repeat pass coherence. One study [15] concluded that the maximum interval length for coherent SAS CD will be between hours and a few days for sonar frequencies above 30 kHz.

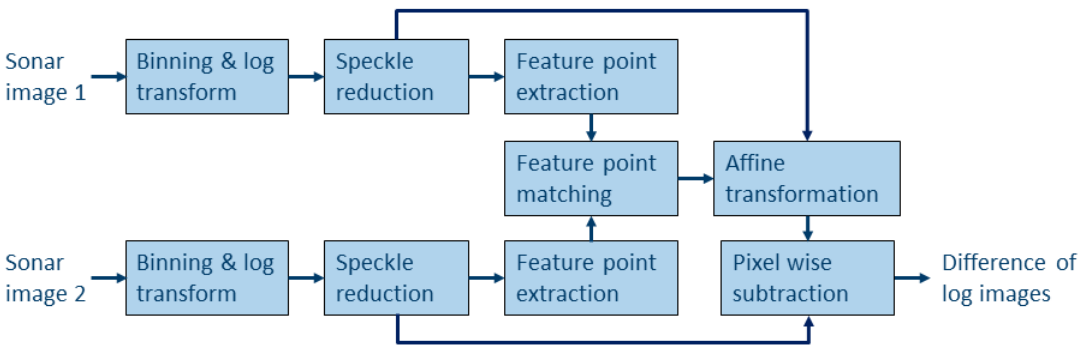

Fig. 1: Incoherent, image level change detection processing used in this study.

Incoherent, image level CD can only detect changes that are visible in the magnitude sonar images, but it also has weaker requirements, as coregistration accuracy at most needs to equal the pixel size. It is also more tolerant to sensor track offsets, temporal seafloor changes and signal propagation variations. One implementation has demonstrated promising capabilities for object detection within dense clutter [11], and was also used in this study. The processing chain is shown in Fig. 1 and is shortly described below.

The back-projection algorithm is initially used to create single pass SAS images onto a ground plane with grid size 2cm x 2cm, which is slightly smaller than the theoretical resolution of HISAS 1030. Corresponding images from repeated passes cover the same geographical region based on vehicle navigation data, with a typical offset of a few meters. The images are down sampled by averaging to grid size 4cm x 4cm to obtain independent neighbor pixels. A logarithmic transform is applied to emphasize sonar highlights and shadows more equally. After image speckle filtering, the SURF algorithm [16] is used to extract and describe feature points in both SAS images. Matching point pairs are identified and used to estimate the parameters for affine transformation of the reference image onto the pixels of the repeat pass image. After this spatial transformation, pixel-wise subtraction produces the difference image. This image can be thresholded or further processed to detect specific response change pattern dependent on the application.

3. EXPERIMENTS

The sensor data in this study was recorded at two different sites in Norwegian coastal waters using various HUGIN AUVs equipped with HISAS 1030 sonar (centre frequency 100 kHz) and TileCam camera system. The small scale sonar response of both areas was expected to change over time, as the seabottoms consist of grained deposits and are affected by biological activity. TileCam optical images from both sites are shown in Fig. 2. Both sites were repeatedly surveyed, with time intervals ranging from a few days between the first two surveys, up to a year or more between the first and last surveys (Table 1). Different vehicles, and thus sensor units, were used in the first and last survey at both sites. The surveys in January 2009 were performed by Kongsberg Maritime with their vessel *Simrad Echo*, while the other surveys were conducted by the Royal Norwegian Navy Mine Warfare Service and FFI with the naval vessels *KNM Karmøy* and *KNM Tyr*.

Fig. 2: TileCam optical images of the two test sites. Left: Mud seafloor with debris and animal induced pits at site A (image size: 3.6m x 2.4m). Right: Concrete cube on textured sand seafloor at site B (image size: 6.5m x 4.3m, recorded February 2011).

Test site	*A* *mud seafloor, depth around 70 m*			*B* *sand seafloor, depth around 25 m*		
Survey date	*6-1-2009*	*8-1-2009*	*16-6-2012*	*10-2-2011*	*14-2-2011*	*6-2-2012*
Vehicle	*HUGIN 1000-MR*		*HUGIN 1000-HUS*	*HUGIN 1000-KM3*		*HUGIN 1000-MR*
Targets	*None*			*None*	*Concrete cube*	*Cube + GRP shell*

Table 1: Summary of SAS surveys used in this study.

The seafloor in site A is fairly flat and consists of soft deposits (presumably mud) with various scattered debris and isolated rockfields. Water depth is around 70 m, and TileCam imagery has documented biological bottom activity due to e.g. crawfish, sea cucumbers and annelids. No targets were deployed between the surveys in this area, as the aim was merely to study temporal background changes. The seafloor in site B is also fairly flat and consists of sand with pebbles, shell fish and sparse kelp. Water depth is around 25 m. A concrete cube (40 x 40 x 40 cm) was permanently deployed at the seafloor just before the second survey, and a thin, cup-shaped shell (height 47 cm, top diameter 1 m) of glass-reinforced plastic (GRP) was temporarily deployed during the third survey.

4. RESULTS AND DISCUSSION

SAS images covering 120m x 120m were produced for all surveys from both test sites. Despeckled sub images measuring 30m x 30m are presented in Fig. 3 (site A) and Fig. 4 (site B) together with the difference images. Dynamic range is 45 decibels for all images.

The images recorded two days apart at site A (Fig. 3a,b) look identical, with similar responses from debris objects, seafloor drag marks and pits. The most striking difference is perhaps the large cross track shift of almost 15 meters, as evident by the scale on the horizontal axis. This far exceeds the typical position error for the case of repeated survey lines with aided navigation on HUGIN AUVs. In spite of this large offset, the response dynamics have been almost perfectly eliminated in the difference image (Fig. 3d). At short ranges, however, this change in viewing geometry alters object shadow lengths which leave a small residual in the difference image. Similarly, shadows may be only partly suppressed for input images with different SNRs, as the shadow contrast to the surrounding seafloor will differ. The repeat pass image for a time separation of almost 3½ years (Fig. 3c) is obviously different from the two previous images. The drag scars are heavily eroded and many new pits have appeared. The corresponding difference images (Fig. 3 e,f) thus contain a lot of pit responses as well as some faint, inverted drag scars, but the stationary debris responses are effectively filtered out. The similarity of these two difference images strongly indicates that the method produces consistent results when the sensing conditions are compatible.

The ability to produce a difference image where mutual variations are suppressed, while temporal changes are presented as they appear in the single pass images, is one significant advantage of the proposed CD method. It enables the detection of general changes rather than only specific features and makes it intuitive to interpret the changes, e.g. to differentiate between appeared and disappeared objects. Dedicated processing of the decluttered difference image can be used to extract only the relevant changes.

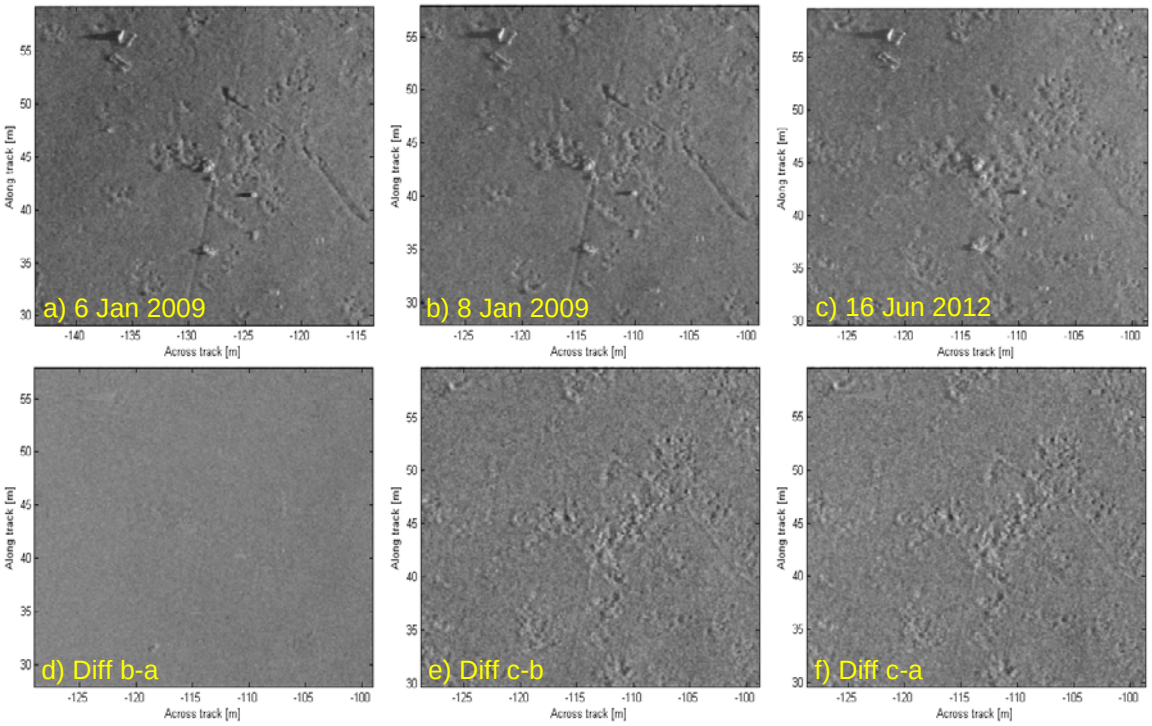

Fig. 3: SAS image snippets (30m x 30m) from site A surveys (top row) and their temporal difference images (bottom row).

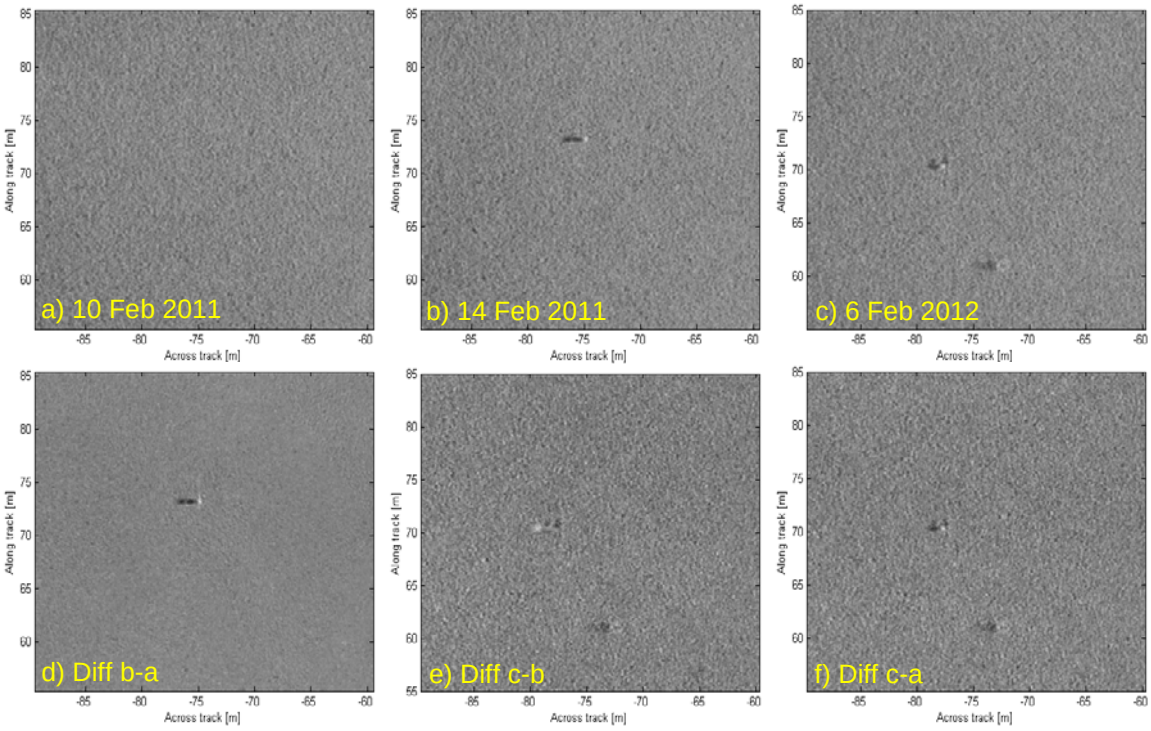

Fig. 4: SAS image snippets (30m x 30m) from site B surveys (top row) and their temporal difference images (bottom row). A concrete cube was deployed before the second survey and a glass fiber shell before the third survey.

The reference image from site B (Fig. 4a) does not contain any large bottom features, but has an overall small scale texture due to the spatial distribution of pebbles, shell fish, etc. This background texture is accurately replicated in the repeat pass image recorded four days later (Fig. 4b), albeit with a minor offset caused by relative errors in the vehicle

navigation solutions. When this offset is corrected by feature point coregistration, the corresponding difference image (Fig. 4d) has an almost perfectly smooth background, thus enhancing the concrete cube response. The image recorded almost one year later (Fig. 4c) reveals a shortened cube shadow, suggesting that the object is now partly immersed into the seafloor. The newly deployed, semi-transparent glass fiber shell is also visible as a weak highlight and shadow in the lower part the image. Comparison with the reference image shows that the images' background details now are completely different. For two uncorrelated backgrounds, the fluctuations (variances) accumulate when the images are subtracted, rather than diminish. The resulting difference image (Fig. 4f) thus has somewhat more background fluctuations than either of the two input images. Although both targets are visible in the difference image, their responses are not enhanced compared with the single pass image. The difference image between the two last surveys (Fig. 4e) shows that the cube response now is suppressed, except for the inconsistent shadow tail and a couple of hollows (possibly due to scouring), while the shell response is preserved.

The amount of stationary background features obviously affects the feature point processing. For the full 120m x 120m images, more than 3000 matching point pairs were found with four days interval for site B, while only 16 and 11 matches were found with one year intervals. The shortage of permanent clutter or other features which can be used for accurate coregistration makes site B a challenging case for the proposed CD method. It also makes it feasible to use decision level methods. However, the results show that the proposed method works even for a seafloor where stationary sonar features are scarce. The method thus holds potential as a general method that can be applied both in cluttered and benign areas. The cases where coregistration does not work due to fundamental background changes, e.g. a flat seafloor that has become rippled after a storm, can be automatically flagged by their lack of feature point matches.

5. CONCLUSIONS

This study has demonstrated that incoherent, image level change detection can suppress stationary clutter (e.g. rocks and debris) in SAS imagery, even for time intervals of one year or more and with up to 15 m survey track offset. The method produces a difference image which may prove useful in many SAS change detection applications, including mine hunting, pipeline inspection and environmental monitoring. The results have been obtained on sonar data recorded at two sites in Norwegian coastal waters, constituting two different seafloor environments.

As expected, the temporal background changes caused by biological and geophysical processes on the seafloor increase with survey interval length. For intervals of a few days, the difference image backgrounds were almost homogeneous, while separation of a year or more yielded considerable small scale variations.

6. ACKNOWLEDGEMENTS

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